

Chapter 10

Quantitative Disaccharide Profiling of Glycosaminoglycans from Two Different Preparations by PMP and Deuterated PMP Labeling

Meng Zhang, Yi An, and Lijuan Zhang

Abstract

Glycosaminoglycan (GAG) fine structures from the same animal cells and tissues are controlled not only by the biosynthetic and metabolic enzymes but also by other environmental factors, such as chemicals, growth factors, nutritional factors, and isolation procedures. To facilitate direct quantitative comparison of disaccharide compositions from different GAG preparations, several stable isotope labeling strategies have been developed. In this report, 1-phenyl-3-methyl-5-pyrazolone (PMP) and deuterated d5-PMP are used for differential disaccharide labeling and profiling of chondroitin sulfate GAG by high performance liquid chromatography (HPLC) coupled with mass spectrometry (MS).

Key words Glycosaminoglycan, Chondroitin sulfate, Disaccharide profiling, 1-Phenyl-3-methyl-5-pyrazolone (PMP), Deuterated d5-PMP, High performance liquid chromatography, Mass spectrometry

1 Introduction

Radioisotopic or UV/fluorescence labeling of glycosaminoglycan (GAG) di- or oligosaccharides generated by either enzymatic or chemical cleavage methods is generally used for establishing the structure–function relationship of GAGs [1, 2].

1-phenyl-3-methyl-5-pyrazolone (PMP) was earlier reported to facilitate monosaccharide composition analysis of glycoproteins by Honda et al. in 1989 [3]. Due to its strong ultraviolet (UV) absorbing ability and hydrophobicity allowing for high-resolution chromatographic separation and high-sensitivity spectrometric detection, PMP labeling subsequently becomes a common method [4] for monosaccharide compositional or structural analyses of different glycans and quality assessment of glycoside sweeteners [5–11]. Furthermore, PMP and deuterated d5-PMP have been used for structural comparisons of *N*- and *O*-

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linked glycans released from glycoproteins [12–14] due to its simplicity and reliability when coupled with HPLC-MS analysis.

In recent years, several stable isotope labeling strategies, but not PMP and d5-PMP, have been developed to perform pairwise di- and oligosaccharide structural comparisons of GAGs [15–20]. The isotope labeling has no effect on retention time upon HPLC separation but can be discriminated by a mass detector, and the method sensitivity is only limited by the type of mass spectrometers available. Thus, we first used the PMP strategy to analyze monosaccharide compositions of GAGs and glycans from different sources [9, 21–23]. We subsequently synthesized the d5-PMP and developed the PMP and d5-PMP strategy for enzymatically digested disaccharide composition analysis of GAGs either from commercial sources or isolated from drug-treated animal cell and tissue samples [24–26].

As a demo for this method, the PMP- and d5-PMP-labeled disaccharides of GAGs were well separated on a C18 reversed-phase column without any need for conventional ion-pairing agents [20, 27] (Fig. 1). The obtained MS data feature a permanent 10-Da mass difference between the DMP and d5-PMP-labeled disaccharides, allowing complete discrimination and comparative quantification of these isotopically labeled GAG disaccharides (Fig. 2).

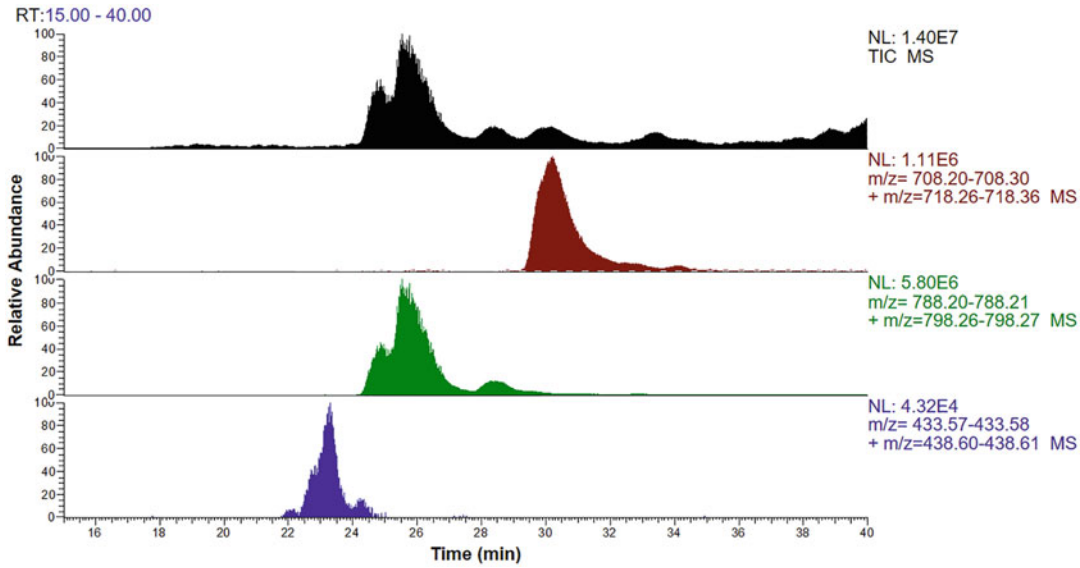


Fig. 1 Extracted ion chromatograms for all PMP-labeled disaccharides from the full-scan HPLC-MS. The d5-PMP-labeled disaccharides are co-eluted at the same intensity but the overlaid extracted ion chromatograms are not shown for simplicity (*see* **Note 4**)

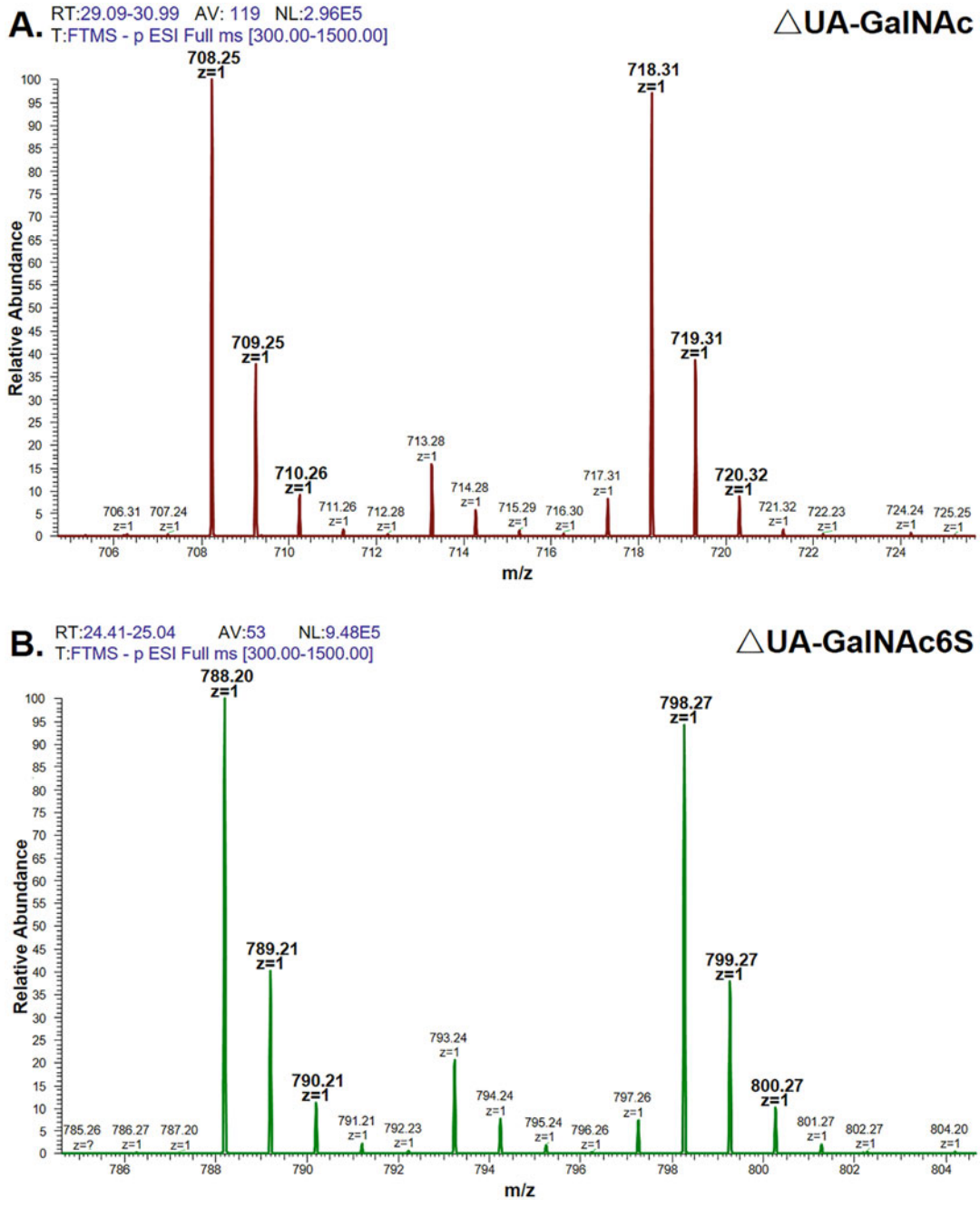


Fig. 2 Mass detection of PMP and d5-PMP-labeled disaccharides of chondroitin sulfate A (*see* **Note 4**). (**a**) ΔUA-GalNAc ($m/z = 708$ with 2 × PMP and 788 with 2 × d5-PMP). (**b**) ΔUA-GalNAc6S ($m/z = 788$ with 2 × PMP and 798 with 2 × d5-PMP). (**c**) ΔUA-GalNAc4S ($m/z = 788$ with 2 × PMP and 798 with 2 × d5-PMP). (**d**) ΔUA2S-GalNAc ($m/z = 788$ with 2 × PMP and 798 with 2 × d5-PMP). (**e**) ΔUA-GalNAc4S6S ($m/z = 433.5$ with 2 × PMP and 438.5 with 2 × d5-PMP, $z = 2$)

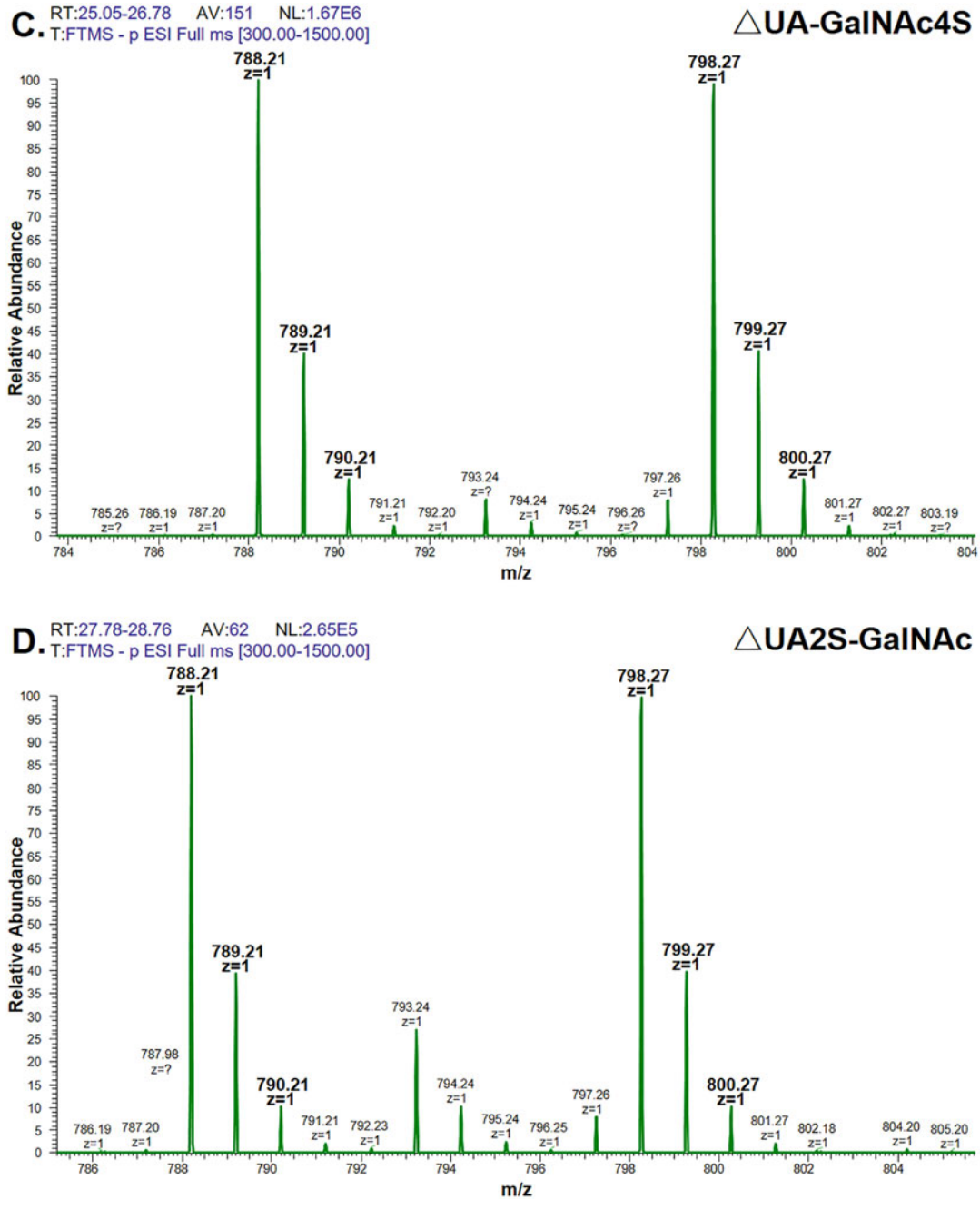


Fig. 2 (continued)

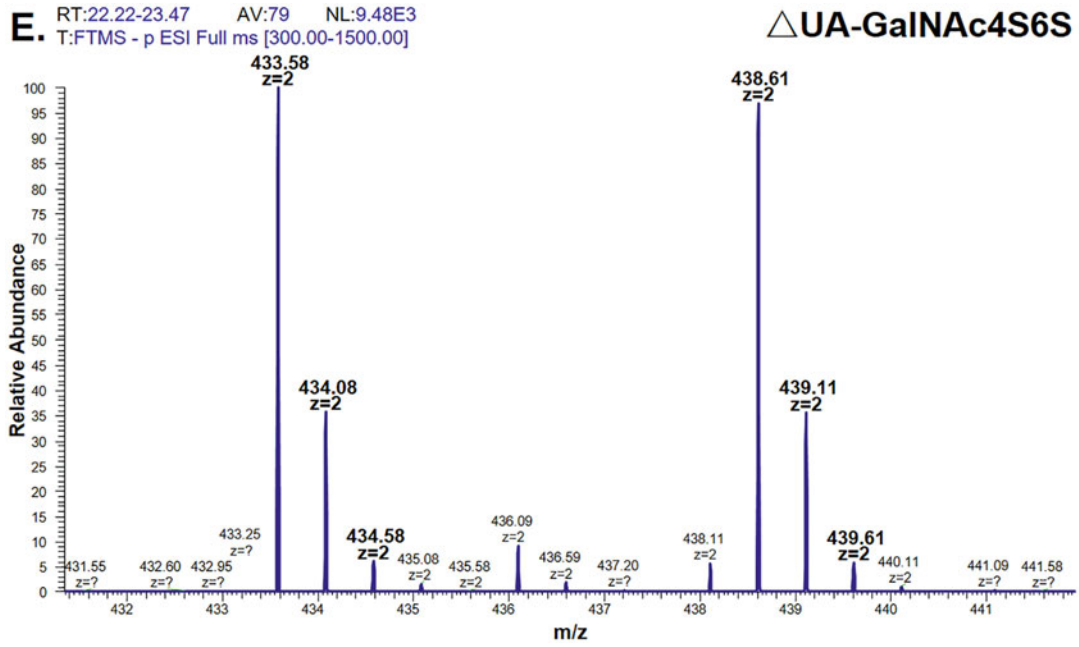


Fig. 2 (continued)

2 Materials

1. Chondroitin sulfate A.
2. Chondroitinase ABC.
3. Chondroitin sulfate disaccharide standards.
4. HPLC-grade acetonitrile.
5. 1-Phenyl-3-methyl-5-pyrazolone (PMP).
6. Deuterated d5-aniline.
7. Deionized water (18.2 MΩ).

3 Methods

3.1 Synthesis of d5-PMP

1. Add 5 g of d5-aniline to 25 mL of 6 mol/L hydrochloric acid (HCl) solution and stir for 30 min in an ice bath. $NaNO_2$ aqueous solution (6.5 mL, 35% w/v) is added dropwise and stirred in an ice bath for 1 h.
2. Add 50 mL 0.2 mol/L $SnCl_2$ solution dropwise at a constant speed under freezing point, and stir for 3–5 h.
3. Extract the product with dichloromethane, the organic layer is collected and dried over anhydrous magnesium sulfate to

obtain d5-phenylhydrazine. The product is filtered with suction force and rinsed with dichloromethane. Dissolve the product in 50 mL water, adjust the pH to 10 with aqueous NaOH solution, and stir until dissolved.

4. Ethyl acetoacetate is added to the d5-phenylhydrazine solution at a volume ratio of 1:0.778, and the reaction is carried out at 50 °C in water bath by using an ultrasonic instrument. After monitoring the completion of the reaction using a TLC plate, the reaction product is dissolved in methanol, filtered, and evaporated to dryness to obtain solid d5-PMP.

3.2 Enzymatic Digestion of Chondroitin Sulfate A

1. Chondroitin sulfate A is dissolved in deionized water (18.2 MΩ cm, 25 °C) to prepare 10 mg/mL chondroitin sulfate A stock solution.
2. Add 10 μL chondroitin sulfate A stock solution into 200 μL 100 mmol/L ammonium acetate buffer solution (pH = 7.0), then add 5 μL of 10 mU/mL chondroitinase ABC and incubate at 25 °C (*see* **Note 1**).
3. Monitor the enzymatic digestion reaction by observing UV absorption at 232 nm in a microplate reader. After the UV absorption intensity is stable, lyophilize the digested chondroitin sulfate A sample.

3.3 Derivatization of Disaccharides with PMP and d5-PMP

1. The derivatization is modified from the protocol of Honda [3, 28], in that the digested chondroitin sulfate A sample is dissolved in 200 μL deionized water and aliquoted into two parts before 30 μL PMP or d5-PMP methanol solution (0.0871 mg/mL) is added to each aliquot in Eppendorf tube.
2. Add 5 μL ammonia solution to each sample to adjust the pH to 11–13 (*see* **Note 2**), tightly seal the Eppendorf tube to allow the reaction to occur at 70 °C for 60 min.
3. Reaction products are swirled with 500 μL chloroform to extract the excess unreacted PMP (*see* **Note 3**) and followed by centrifuging at 15,000 × *g*/min for 5 min. The extraction is repeated three times.
4. The aqueous layer is collected carefully before HPLC analysis.

3.4 Detection by HPLC-MS

1. The settings for HPLC analysis are as follows: 0.4 μL sample is injected and running time is 80 min at 0.01 mL/min flow rate. Samples are separated by C18 reversed-phase column (ZOBARX, SB-C18, 2.1 × 150 mm, 1.8 μm, Agilent) at 30 °C. UV absorbance at 254 nm is monitored and recorded.
2. The settings of the mass spectrometer are: the drying gas flow rate is 12 L/min and a nebulizer pressure is 50 psi. The drying gas temperature is set at 350 °C. The value of the fragmentor is

set at 100 V with capillary voltage at 4000 V in positive ion scan mode and 3500 V in negative ion scan mode.

3. Mobile gradient: phase A (acetonitrile): 0–50 min (17% → 23%), 50–65 min (23% → 23%), 65–65.1 min (23% → 17%), 65.1–80 min (17% → 17%); phase B (0.01 mol/L ammonium acetate solution, pH = 4.5): 0–50 min(83% → 77%), 50–65 min (77% → 77%), 65–65.1 min (77% → 83%), 65.1–80 min (83% → 83%).

4 Notes

1. Chondroitinase ABC is temperature sensitive. The stock solution of chondroitinase ABC should be stored at −80 °C and thawed on ice just before use.
2. PMP derivatization only occurs at alkaline conditions. However, excessive alkali would lead to the reaction failure. The suitable pH range should be between11 and 13, which can be monitored by taking 1 μL aliquot on to a pH test paper strip.
3. Both the volume of chloroform and the length of extraction time affect the efficiency of extraction. When the extraction volumes are below 500 μL, substantial amount of unreacted PMP is still present in the final derivatizing product. When extracted more than three times, apparent loss of PMP- and d5-PMP disaccharides is observed.
4. The PMP/d5-PMP–labeled disaccharide analysis described in Figs. 1 and 2 showed our efforts in developing a high resolution/mass accuracy reverse-phase HPLC-MS platform for pairwise quantitative disaccharide profiling from equal amount of commercially available chondroitin sulfate A: each PMP/d5MMP-labeled disaccharides that can be identified and quantified based on the MS signal intensity based on the molecular weight difference (Δ10 Da for z1 ions in Fig. 2a–d and Δ5 Da for z2 ion in Fig. 2e). Most importantly, this technique has been successfully applied to monitor the changes in disaccharide compositions in both heparan sulfate and chondroitin sulfate using the GAGs isolated from either animal cell or tissues after treating with different drugs [24–26].

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